aws

Architecture Diagrams

AI/ML Based Intelligent Email Responder



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AI/ML Based Intelligent Email Responder: Architecture Diagrams

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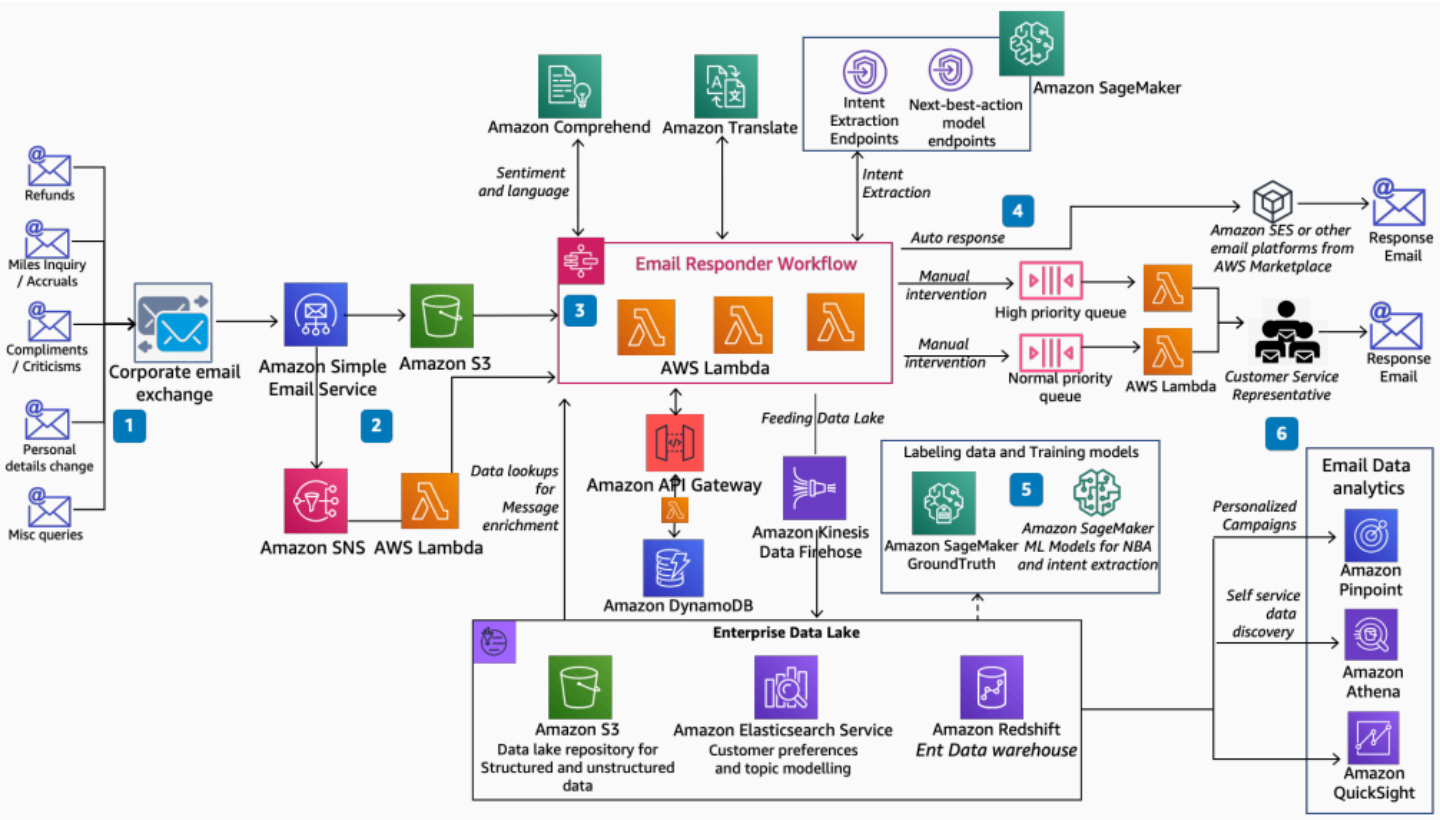
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AI/ML Based Intelligent Email Responder

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This architecture shows how to improve customer experience and agent productivity. With Amazon Simple Email Service (Amazon SES) and AWS ML services, you can comprehend intent and identify next-best-action for email responses.

AI/ML Based Intelligent Email Responder



The following steps describe the architecture:

1. Amazon SES captures emails received at the corporate email exchange.
2. Amazon SES writes the raw payload to Amazon Simple Storage Service and triggers a notification through Amazon Simple Notification Service. An AWS Lambda function captures this notification and triggers the email responder workflow.
3. The email responder workflow uses Amazon Comprehend for language, sentiment, and keyword extraction. It uses Amazon Translate for language translation. It calls intent extraction and next-best-action model endpoints deployed on Amazon SageMaker AI for better routing. It performs

data lookups from the enterprise data lake. It updates the transactional data stores by using Amazon Kinesis Data Firehose.

4. Based on confidence and completeness, the workflow either sends a response through Amazon SES or routes the email to the right queue for an agent.
5. From the data collected, you continuously improve the intent extraction and next-best-action models by using SageMaker AI Ground Truth and SageMaker AI.
6. You can perform personalized email campaigns, additional email analytics, and reporting by using Amazon Pinpoint, Amazon Athena, and Amazon Quick Sight.

Further reading

For additional information, refer to the following resources:

- AWS Architecture Icons
- AWS Architecture Center
- AWS Well-Architected
- Amazon SageMaker AI product page

Diagram history

To be notified about updates to this reference architecture diagram, subscribe to the RSS feed.

Change	Description	Date
Initial publication	Reference architecture diagram first published.	March 10, 2021

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